

### Multiple Choice Questions

Each question is worth 1 mark. No partial credit.

- Q1.** Many scholars and policy makers are interested in the proportion of homeless people who are mentally ill. Homeless individuals were sampled from those who received medical attention at clinics in the Health Care for the Homeless (HCH) project. Which best describes the sampling frame in this study?
- (A) All homeless individuals in the geographic region covered by the study
  - (B) All homeless individuals who had contact with healthcare services during the study period
  - (C) All homeless individuals who received care at HCH clinics during the study period
  - (D) All individuals who received medical services and completed the survey or assessment
- Q2.** In the survey from Q4, it is argued that selection bias is not a serious problem because the clinics were easily accessible to homeless individuals and because the demographic profiles (age, sex, race) of HCH clients were similar to those of the general homeless population in each city in the sample. Do you agree with this argument?
- (A) Yes, because accessibility and similar demographic distributions imply that the sample is effectively representative of the target population
  - (B) Yes, because any study that includes a large number of participants automatically eliminates selection bias concerns
  - (C) No, because accessibility and similar demographic distributions does not guarantee that all subgroups of the homeless population have equal probability of inclusion
  - (D) No, because selection bias is only a concern when response rates are extremely low
- Q3.** A researcher wants to study all full-time undergraduate students at UofT in Fall 2026. They use the registrar's list of active student emails to select participants. What is the **sampling frame**?
- (A) All full-time undergraduates at UofT in Fall 2026
  - (B) The registrar's list of active student emails
  - (C) Each individual student email selected
  - (D) Students who respond to the survey
- Q4.** A telephone survey uses landline numbers to estimate average income in a city. Which source of selection bias is most relevant?
- (A) Overcoverage, because some numbers are listed multiple times
  - (B) Undercoverage, because people without landlines (often lower-income) are excluded
  - (C) Nonresponse, because some people hang up
  - (D) Convenience sampling, because the interviewer chose who to call

- Q5.** A researcher surveys adolescents willing to speak with investigators about AIDS to estimate how often teenagers discuss AIDS with parents. Why is this problematic?
- (A) The sample size is too small
  - (B) Teens willing to talk to investigators about AIDS are also likely more willing to talk to parents, inflating the estimate
  - (C) The survey uses closed questions that limit responses
  - (D) The target population is not defined clearly enough
- Q6.** A provincial health survey is conducted by mailing questionnaires to addresses drawn from a voter registration list. Individuals experiencing homelessness are not on voter rolls and have no chance of being included. What type of selection bias does this describe?
- (A) Nonresponse bias
  - (B) Convenience sampling
  - (C) Overcoverage
  - (D) Undercoverage
- Q7.** A survey of radio listening habits targets adults aged 18+, but some interviewers accidentally include 15–17 year olds when conducting phone interviews because they were not given specific age-screening instructions. What type of selection bias has occurred?
- (A) Overcoverage
  - (B) Undercoverage
  - (C) Nonresponse bias
  - (D) Self-selection bias
- Q8.** A mail-in survey on pension satisfaction is sent to 5,000 retired government employees. Only 820 return the questionnaire. A follow-up study finds that those who did not respond had significantly lower pension incomes on average. Which type of selection bias best explains the threat to validity here?
- (A) Convenience sampling
  - (B) Undercoverage
  - (C) Nonresponse bias
  - (D) Overcoverage

- Q9.** A public health agency mails a survey about alcohol consumption to 10,000 randomly selected adults. Only 2,200 respond. A researcher later finds that heavy drinkers were significantly less likely to return the questionnaire. Which type of selection bias explains why alcohol consumption estimates will be biased downward?
- (A) Undercoverage
  - (B) Overcoverage
  - (C) Convenience sampling
  - (D) Nonresponse bias
- Q10.** A researcher posts a voluntary survey link in a private Facebook group for Toronto dog owners to estimate average weekly pet care spending among all Toronto pet owners. Only group members who happen to see the post and choose to click the link respond.
- (A) Undercoverage (only Facebook dog-owner group members are reachable) and self-selection (only motivated members respond)
  - (B) Overcoverage (the group is too large) and nonresponse (many members didn't reply)
  - (C) Convenience sampling (Facebook is easy to use) and overcoverage (group includes non-pet-owners)
  - (D) Nonresponse (low reply rate) and undercoverage (cat owners excluded from the group)
- Q11.** A university sends a course evaluation survey by email to all 229 enrolled students. Only 51 students complete it (a 22% response rate). Students with very positive or very negative experiences are most likely to take the time to respond.
- (A) Undercoverage (some students don't check email) and overcoverage (TAs are accidentally included)
  - (B) Nonresponse bias (78% didn't reply) and self-selection (those with strong feelings are more likely to respond)
  - (C) Convenience sampling (only in-person students are reachable) and undercoverage (part-time students excluded)
  - (D) Overcoverage (alumni also receive the email) and nonresponse (some students ignored it)
- Q12.** A survey asks: "Have you smoked at least 100 cigarettes in your entire life?" Some respondents include marijuana cigarettes; others exclude partially-smoked ones. This is an example of:
- (A) Selection bias due to self-selection
  - (B) Measurement error caused by question ambiguity
  - (C) Nonresponse bias
  - (D) Overcoverage in the sampling frame

- Q13.** A survey asks: “Given that most experts agree this policy is harmful, do you support or oppose it?” This question violates which guideline?
- (A) It uses an agree/disagree format instead of a forced-choice format
  - (B) It is a leading question — the preamble nudges respondents toward opposition
  - (C) It is double-barreled
  - (D) It asks about a concept too vague to measure
- Q14.** Which revision best improves the question “The instructor created an atmosphere conducive to my learning” (rated on a 1–5 scale)?
- (A) “The instructor was a very good teacher overall” (1–5 scale)
  - (B) “Did the instructor encourage students to ask questions during class?” (Yes / No)
  - (C) “Did you enjoy the atmosphere in this class?” (1–5 scale)
  - (D) “Do you agree that the atmosphere was good and the instructor was helpful?”
- Q15.** A poll reports a  $\pm 3\%$  margin of error (half width of confidence interval). A colleague notes the response rate was only 25%. Which statement is most accurate?
- (A) The  $\pm 3\%$  margin fully captures all error in the poll
  - (B) The  $\pm 3\%$  captures sampling error only; the low response rate introduces non-sampling error
  - (C) Both sampling and non-sampling errors are included in the  $\pm 3\%$
  - (D) Non-sampling error only occurs when the sample is very small
- Q16.** Which is a genuine advantage of a well-designed sample over a census?
- (A) A sample has no sampling error, unlike a census
  - (B) Data collection can be more carefully managed, often producing more accurate measurements than a rushed census
  - (C) A sample covers every member of the population
  - (D) A sample eliminates all nonresponse
- Q17.** A researcher studying injection drug users recruits initial participants, who then refer others in their social network. This is:
- (A) Stratified sampling
  - (B) Snowball sampling
  - (C) Systematic sampling
  - (D) Cluster sampling

- Q18.** A population is  $\mathcal{U} = \{1, 2, 3, 4\}$ . A sampling scheme assigns  $P(\{1, 2\}) = 1/3$ ,  $P(\{1, 3\}) = 1/6$ ,  $P(\{3, 4\}) = 1/2$ , and zero to all other size-2 samples. What is  $\pi_3$ , the inclusion probability of unit 3?
- (A)  $1/6$   
 (B)  $1/2$   
 (C)  $2/3$   
 (D)  $5/6$
- Q19.** A population has  $N = 20$  units. You draw an SRS of size  $n = 4$  without replacement. Given that unit 7 is already in the sample, what is the conditional probability that unit 12 is also in the sample?
- (A)  $4/20$   
 (B)  $3/19$   
 (C)  $1/5$   
 (D)  $3/20$
- Q20.** Population values are  $y_1 = 2, y_2 = 4, y_3 = 6, y_4 = 8$  ( $N = 4$ ). An SRS without replacement of size  $n = 2$  is drawn. Consider the estimator  $\hat{t} = \frac{N}{n} \sum_{i \in \mathcal{S}} y_i$  for the population total  $t$ . Under SRS, which of the following is true?
- (A) Unbiased;  $\text{Var}(\hat{t}) = 0$   
 (B) Unbiased;  $\text{Var}(\hat{t}) = 27$   
 (C) Biased upward;  $\text{Var}(\hat{t}) = 27$   
 (D) Unbiased;  $\text{Var}(\hat{t}) = 2$
- Q21.** A population has  $N = 6$  units labelled  $\{1, 2, 3, 4, 5, 6\}$ . An SRS without replacement of size  $n = 2$  is drawn. What is the probability that the sample contains at least one even-numbered unit?
- (A)  $13/15$   
 (B)  $4/5$   
 (C)  $2/3$   
 (D)  $14/15$
- Q22.** In the probability sampling framework, the  $y_i$  values are treated as fixed constants, while the randomness comes entirely from the sampling mechanism. Which consequence follows directly from this perspective?
- (A) We must assume the  $y_i$  are drawn from a probability distribution to make inferences  
 (B) The sampling distribution of an estimator arises from the random selection of units, not from modelling the  $y_i$  as random  
 (C) Bias and variance of an estimator depend on the assumed distribution of  $y_i$   
 (D) Confidence intervals require the Central Limit Theorem applied to  $y_i$

- Q23.** Under a Simple Random Sampling without replacement (SRS) design of size  $n$  from a finite population of size  $N$ , let  $I_i$  be an indicator random variable that equals 1 if unit  $i$  is included in the sample and 0 otherwise. What is the expected value  $E[I_i]$ ?
- (A)  $1/N$
  - (B)  $1 - n/N$
  - (C)  $1/n$
  - (D)  $n/N$
- Q24.** Which of the following statements are true?
- (A) Strata should be similar to one another (ie. they should be homogeneous).
  - (B) None of the other options are correct.
  - (C) Clusters should not be similar to one another (ie. they should be heterogeneous).
  - (D) Strata should not be similar to one another (ie. they should be heterogeneous).
- Q25.** In the probability sampling framework, the  $y_i$  values are treated as fixed constants, while the randomness comes entirely from the sampling mechanism. Which consequence follows directly from this perspective?
- (A) We must assume the  $y_i$  are drawn from a probability distribution to make inferences.
  - (B) The sampling distribution of an estimator arises from the random selection of units, not from modelling the  $y_i$  as random.
  - (C) Bias and variance of an estimator depend on the assumed distribution of  $y_i$ .
  - (D) Confidence intervals require the Central Limit Theorem applied to  $y_i$ .
- Q26.** A researcher wants to estimate the proportion of wild salmon in a river that carry a particular parasite. She argues that she could use simple random sampling by tagging every salmon. A colleague objects that SRS is not appropriate here. What is the most fundamental reason the colleague is right?
- (A) The population proportion is too close to zero for SRS to be valid.
  - (B) A complete sampling frame of individual salmon cannot be constructed, which is a prerequisite for SRS.
  - (C) SRS requires the population to be normally distributed.
  - (D) The finite population correction cannot be computed for animal populations.
- Q27.** The sample mean  $\bar{y}_S$  is an unbiased estimator of the population mean  $\bar{y}_U$  under SRS. Which derivation step is essential in showing this?
- (A) Using the Central Limit Theorem to approximate the distribution of  $\bar{y}_S$ .
  - (B) Applying  $E[I_i] = n/N$  so that  $E[\bar{y}_S] = \frac{1}{n} \sum_{i=1}^N \frac{n}{N} y_i = \bar{y}_U$ .
  - (C) Assuming  $y_i$  are i.i.d. random variables.
  - (D) Showing that the sample variance  $s^2$  equals  $\sigma_U^2$ .

**Q28.** For very skewed populations, Sugden et al. (2000) recommend a minimum sample size of

$$n_{\min} = 28 + 25 \left[ \frac{\sum_{i=1}^N (y_i - \bar{y}_U)^3}{N \sigma_U^3} \right]^2.$$

What does the bracketed expression measure, and why does it affect the required sample size?

- (A) It measures population kurtosis; high kurtosis increases the variance.
- (B) It measures the population skewness; high skewness slows convergence to normality in the CLT, requiring larger  $n$  for the normal approximation to be accurate.
- (C) It measures the sampling fraction; larger fractions require larger samples.
- (D) It measures the bias of  $\bar{y}_S$ ; a correction is needed when bias is nonzero.

**Q29.** Suppose  $n/N = 0.02$ . A colleague proposes ignoring the finite population correction. Which argument best justifies this decision?

- (A) The fpc equals  $1 - 0.02 = 0.98$ , changing the standard error by less than 1%, which is negligible for practical inference.
- (B) The fpc can always be ignored when the population is finite.
- (C) Ignoring the fpc eliminates the need for a sampling frame.
- (D) The fpc only applies to proportion estimators, not means or totals.

**Q30.** Under SRS, the covariance between indicators  $I_i$  and  $I_j$  ( $i \neq j$ ) is negative. Which feature of the sampling design directly causes this?

- (A) Units are selected with unequal probabilities.
- (B) Selecting unit  $i$  without replacement uses one of the  $n$  available slots, reducing the probability that unit  $j$  is subsequently selected.
- (C) The population values  $y_i$  and  $y_j$  are negatively correlated.
- (D) The estimator  $\bar{y}_S$  is biased when covariances are nonzero.

**Q31.** The population proportion estimator  $\hat{p}_S$  is a special case of the sample mean. Which of the following correctly states its relationship to  $\bar{y}_S$ ?

- (A)  $\hat{p}_S = \bar{y}_S$  only when  $y_i \in \{0, 1\}$  for all  $i$ .
- (B)  $\hat{p}_S = \bar{y}_S^2$ .
- (C)  $\hat{p}_S = N \bar{y}_S$ .
- (D)  $\hat{p}_S$  and  $\bar{y}_S$  share the same formula only when the population is symmetric.

**Q32.** A design is called *self-weighted* when all units share the same inclusion probability. Which of the following correctly explains why SRS is self-weighted and what this implies for estimation?

- (A) SRS is self-weighted because  $\pi_i = n/N$  for all  $i$ , so every sampled unit carries the same weight  $\omega_i = N/n$ , and the Horvitz–Thompson total estimator reduces to  $N\bar{y}_S$ .
- (B) SRS is self-weighted only when the population is homogeneous.
- (C) SRS is self-weighted because  $\pi_i = 1/N$  for all  $i$ .
- (D) Self-weighting means the estimator has zero variance.

**Q33.** When the population variance  $\sigma_U^2$  is unknown, a  $(1 - \alpha)100\%$  confidence interval for  $\bar{y}_U$  uses  $t_{\alpha/2, n-1}$  rather than  $z_{\alpha/2}$ . Why are  $n - 1$  degrees of freedom used rather than  $n$ ?

- (A) Because we lose one degree of freedom estimating  $\bar{y}_S$  from the sample, which is used in computing  $s^2$ .
- (B) Because the population has  $N - 1$  degrees of freedom.
- (C) Because the finite population correction reduces the effective sample size by one.
- (D) Because  $t_{n-1}$  has heavier tails than  $t_n$ , providing a more conservative interval.

**Q34.** A researcher wants to estimate  $\bar{y}_U$  to within margin  $B$  with confidence  $1 - \alpha$ . The required sample size under SRS is

$$n = \frac{N z_{\alpha/2}^2 \sigma_U^2}{B^2 N + z_{\alpha/2}^2 \sigma_U^2}.$$

As  $N \rightarrow \infty$  with  $B$ ,  $\alpha$ , and  $\sigma_U^2$  held fixed,  $n$  converges to  $z_{\alpha/2}^2 \sigma_U^2 / B^2$ . What does this limiting behaviour reveal?

- (A) For very large populations, a larger and larger sample is always needed.
- (B) The required sample size is ultimately determined by desired precision and population variability, not by population size — the finite population correction becomes irrelevant.
- (C) The formula breaks down for infinite populations and cannot be used.
- (D) Precision improves automatically as  $N$  grows, so no sample is needed.

**Q35.** The stratified estimator of the population mean is defined as  $\bar{y}_{\text{str}} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_h$ . Which of the following correctly describes its bias?

- (A) It is biased upward because larger strata receive more weight.
- (B) It is unbiased, since each  $\bar{y}_h$  is an unbiased estimator of  $\bar{y}_{hU}$  under SRS within each stratum.
- (C) It is biased unless all strata are the same size.
- (D) It is unbiased only when proportional allocation is used.



- Q36.** A firm has  $N_1 = 80$  junior staff with population variance  $\sigma_1^2 = 2$  and  $N_2 = 20$  senior leads with population variance  $\sigma_2^2 = 5$ . The overall population variance is  $\sigma_U^2 = 145$ . A sample of  $n = 10$  is drawn. Under proportional allocation, the variance of  $\bar{y}_{\text{str}}$  is approximately 0.234. What does this reveal about stratified sampling relative to SRS?
- (A) Stratified sampling is less precise because it requires more design decisions.
  - (B) Both designs yield the same variance when sample size is held fixed.
  - (C) Stratified sampling can be substantially more precise than SRS by removing between-stratum variance from the uncertainty calculation.
  - (D) The gain in precision only occurs when strata are equal in size.
- Q37.** The sampling weight for unit  $j$  in stratum  $h$  under stratified sampling is  $\omega_{hj} = N_h/n_h$ . How should this weight be interpreted?
- (A) It is the probability that unit  $j$  is included in the sample.
  - (B) It is the number of population units in stratum  $h$  represented by each sampled unit from that stratum.
  - (C) It ensures the estimator has minimum variance.
  - (D) It is identical across all strata whenever proportional allocation is used.
- Q38.** Under proportional allocation with strata of sizes  $N_1 = 1,600$  and  $N_2 = 400$  and a total sample of  $n = 100$ , what are  $n_1$  and  $n_2$ ?
- (A)  $n_1 = 50$ ,  $n_2 = 50$
  - (B)  $n_1 = 80$ ,  $n_2 = 20$
  - (C)  $n_1 = 75$ ,  $n_2 = 25$
  - (D)  $n_1 = 100$ ,  $n_2 = 0$
- Q39.** Optimal (Neyman) allocation sets  $n_h \propto N_h\sigma_h$ . Compared to proportional allocation, which strata receive relatively more observations under optimal allocation?
- (A) Strata that are large in size regardless of their variability.
  - (B) Strata that are small in size, to compensate for underrepresentation.
  - (C) Strata with high within-stratum variability, since they require more observations to achieve a precise estimate.
  - (D) Strata with low within-stratum variability, since they are easier to estimate.

- Q40.** Proportional allocation is guaranteed to improve precision over SRS when which of the following conditions holds?
- (A) The strata are equal in size.
  - (B) The between-stratum variance exceeds a weighted average of the within-stratum variances.
  - (C) The population size  $N$  is very large.
  - (D) The stratification variable is unrelated to the outcome of interest.
- Q41.** Consider two stratified designs with the same total sample size  $n$  and the same population. Design A uses proportional allocation. Design B stratifies on a variable completely unrelated to the outcome of interest. Which statement is most accurate?
- (A) Design B will always be more precise because it uses more information about the population structure.
  - (B) Design A will always be more precise because proportional allocation minimises variance by construction.
  - (C) Design B will perform approximately as well as SRS, since the between-stratum variance is approximately zero when strata are unrelated to the outcome.
  - (D) Both designs are equivalent because they use the same total sample size.
- Q42.** A population has  $H = 2$  strata with  $N_1 = 900$ ,  $N_2 = 100$ ,  $\sigma_1^2 = 1$ , and  $\sigma_2^2 = 100$ . A sample of  $n = 50$  is available. Under proportional allocation,  $n_1 = 45$  and  $n_2 = 5$ . Under optimal allocation, which of the following is closest to  $n_2$ ?
- (A) 5
  - (B) 14
  - (C) 25
  - (D) 35
- Q43.** Under proportional allocation, the stratified estimator  $\bar{y}_{\text{str}}$  simplifies to the simple average of all sampled observations  $\frac{1}{n} \sum_{h=1}^H \sum_{j \in S_h} y_{hj}$ . Why does this simplification hold?
- (A) Because proportional allocation sets  $n_h = N_h/N$ , making each stratum weight equal to  $1/H$ .
  - (B) Because proportional allocation equates the sampling fraction  $n_h/N_h$  across strata, so every unit has the same inclusion probability  $n/N$  and the same sampling weight  $N/n$ .
  - (C) Because  $\bar{y}_h$  is unbiased for  $\bar{y}_{hU}$  in every stratum under SRS.
  - (D) Because the finite population correction is exactly zero under proportional allocation.

- Q44.** A stratified sample is drawn with  $N_1 = 20$ ,  $n_1 = 10$ ,  $N_2 = 60$ ,  $n_2 = 12$ , and  $N_3 = 420$ ,  $n_3 = 20$ . Which stratum has the largest sampling weight, and what does a large sampling weight imply about representation?
- (A) Stratum 1, with  $\omega_{1j} = 2$ ; each sampled unit represents fewer population units, implying high representation.
  - (B) Stratum 3, with  $\omega_{3j} = 21$ ; each sampled unit represents many population units, implying that stratum 3 is relatively undersampled.
  - (C) Stratum 2, with  $\omega_{2j} = 5$ ; sampling weights are largest for the stratum with the most units sampled.
  - (D) All strata have the same weight because the total sample size is fixed at  $n = 42$ .